1) Top 10 Revenue Products

```
select top 10
p.ProductName , sum(oi.price * oi.Quantity) as TotalRevenue
from Products p
join Order_Items oi
   on oi.ProductID = p.ProductID
group by p.ProductName
order by TotalRevenue DESC
```

Result

```
ProductName        TotalRevenue
---------------------------------
Statement Air      9237298
Stop Max           7055868
Participant Max    6816667
Once X             5198865
Woman Pro          5040840
Baby Ultra         4875882
Trouble Plus       4858863
Something X        4851793
Adult X            4842540
Word X             4817721
```

2) Top Customers by Spending

```
CREATE VIEW Top_Customers_Spending  AS
select top 10
c.CustomerName ,c.Age ,c.Gender  , sum(o.TotalAmount) as Spending
from Customers c
join Orders o
     on o.CustomerID = c.CustomerID
group by c.CustomerName  , c.Age  ,c.Gender
order by sum(o.TotalAmount) DESC
```

Result

```
CustomerName        Age   Gender   Spending
---------------------------------------------
Patricia Thomas     23    male     233057
David Castillo      35    female   216539
Pamela Benitez      62    female   186631
Melissa Smith       40    female   183760
Anna Burton         21    male     180246
Kristy Martinez     48    male     179298
Jamie Martinez      34    male     177945
Jeffrey Bullock     48    male     177490
Maria Douglas       23    male     176936
Sean Morris         55    female   176018
```

3) Best Selling Brands

```
select b.BrandName , sum(oi.Price * oi.Quantity) as Selling
from Brands b
join Products p
   on p.BrandID = b.BrandID
join Order_Items oi
   on oi.ProductID = p.ProductID
group by b.BrandName
order by  selling DESC
```

Result

```
BrandName   Selling
--------------------
Microsoft   135187353
Lenovo      131729853
Acer        128923146
JBL         125653560
OnePlus     125585097
Huawei      125212415
HP          125090440
Bose        114628750
Google      112711012
Apple       111867485
Nikon       110631436
Philips     109695165
Dell        107684216
Sony        106857639
LG          105927417
Samsung     105464751
Canon       103095656
Xiaomi      100648609
Asus        100611769
Panasonic   96703682
```

---

4) Average Order Value

```
select avg(o.TotalAmount) as AverageOrderValue
from orders o
```

Result

```
AverageOrderValue
------------------
2427
```

---

5) Products With Lowest Sales

```
select top 10
p.ProductName ,
sum(oi.Quantity) as TotalSold
,sum(oi.Price * oi.Quantity) as TotalRevenue
from Products p
```

```
 left join Order_Items oi
   on oi.ProductID = p.ProductID
group by p.ProductName
order by TotalSold ASC
```

Result

```
ProductName        TotalSold   TotalRevenue
---------------------------------------------
Sometimes Pro      1272        1865217
Million Max        1303        1923414
Relationship X     1311        1960152
Notice Max         1323        2012566
Player Max         1326        1921069
Outside Plus       1332        1942488
Always Air         1332        1941194
Teacher X          1333        1997685
Ago Air            1335        2011193
Fine Air           1335        2000547
```

---

6) Most Ordered Products

```
select Top 10
p.ProductName , sum(oi.Quantity) as TotalSold
, sum(oi.Price * oi.Quantity ) as TotalRevenue
from Products p
join Order_Items oi
  on oi.ProductID = p.ProductID
group by p.ProductName
order by TotalSold  desc
```

Result

```
ProductName        TotalSold   TotalRevenue
---------------------------------------------
Statement Air      6023        9237298
Stop Max           4614        7055868
Participant Max    4574        6816667
Once X             3408        5198865
Woman Pro          3303        5040840
Baby Ultra         3227        4875882
Our Ultra          3204        4810337
Trouble Plus       3145        4858863
Rest Max           3128        4703286
Way Plus           3125        4634783
```

---

7) Customers With Most Orders

```
select top 10
c.CustomerName , count(o.OrderID) TotalOrders ,
sum(o.TotalAmount) TotalAmount
from Customers c
join orders o
    on o.CustomerID = c.CustomerID
```

```
group by c.CustomerName
order by TotalOrders  DESC
```

Result

```
CustomerName        TotalOrders   TotalAmount
-----------------------------------------------
David Johnson       88            1203079
Michael Johnson     87            1011775
James Smith         83            958731
Matthew Smith       79            946092
Michael Smith       76            816648
Daniel Smith        76            918613
John Johnson        70            741790
Jessica Johnson     62            690620
Michael Garcia      60            650522
Christopher Smith   57            574749
```

8) Average Product Rating

```
SELECT
    p.ProductName,
    AVG(r.Rating) AS AverageRating
FROM Products p
JOIN Reviews r
ON p.ProductID = r.ProductID
GROUP BY p.ProductName
ORDER BY AverageRating DESC;
```

Result

```
ProductName        AverageRating
--------------------------------
Above Plus         3
Above Ultra        3
Across X           3
Activity Pro       3
Actually Air       3
...
```

(معظم المنتجات Average Rating = 3)

---

9) Products Without Reviews

```
select
  p.ProductName
 from Products p
left join Reviews r
     on p.ProductID = r.ProductID
where r.Rating is null
```

Result

```
No Rows Returned
```

يعني **كل المنتجات لها Reviews**.

10) Brand Performance

```
SELECT top 10
    b.BrandName,
    SUM(oi.Quantity) AS TotalSold ,
    Sum(oi.Price * oi.Quantity) as TotalRevenue
FROM Brands b
JOIN Products p
ON b.BrandID = p.BrandID
JOIN Order_Items oi
ON p.ProductID = oi.ProductID
GROUP BY b.BrandName
ORDER BY TotalSold DESC;
```

Result

```
BrandName   TotalSold   TotalRevenue
-------------------------------------
Microsoft   88559       135187353
Lenovo      86030       131729853
Acer        84565       128923146
JBL         83018       125653560
HP          82302       125090440
OnePlus     81992       125585097
Huawei      81685       125212415
Bose        75381       114628750
Google      73631       112711012
Apple       73513       111867485
```

---

11) Revenue Per Customer

```
SELECT
    c.CustomerName,
    SUM(o.TotalAmount) AS Revenue
FROM Customers c
JOIN Orders o
ON c.CustomerID = o.CustomerID
GROUP BY c.CustomerName
ORDER BY Revenue DESC;
```

Result

```
CustomerName        Revenue
-----------------------------
David Johnson       1203079
Michael Johnson     1011775
James Smith         958731
Matthew Smith       946092
Daniel Smith        918613
Michael Smith       816648
John Johnson        741790
Jessica Johnson     690620
```

```
Robert Johnson      669113
Richard Smith       652986
```

---

12) Monthly Sales

```
select
month(o.OrderDate)  , sum(o.TotalAmount) as TotalRevenue
from Orders o
group by  month(o.OrderDate)
order by TotalRevenue DESC
```

Result

```
Month   TotalRevenue
---------------------
7       214204959
10      212920742
12      212700403
8       209873005
1       209824189
9       206698469
11      203080580
2       195205736
3       178429513
6       159172610
5       142144520
4       139654725
```

---

13) Yearly Sales

```
select
year(o.OrderDate)  , sum(o.TotalAmount) as TotalRevenue
from Orders o
group by  year(o.OrderDate)
order by TotalRevenue DESC
```

Result

```
Year   TotalRevenue
-------------------
2025   836544727
2024   832050947
2023   446144814
2026   169168963
```

---

14) Sales by Quarters

```
SELECT
    DATEPART(QUARTER, OrderDate) AS Quarter,
    SUM(TotalAmount) AS TotalRevenue
from  Orders
```

```
group by DATEPART(QUARTER, OrderDate)
order by  Quarter;
```

Result

```
Quarter   TotalRevenue
----------------------
1         583459438
2         440971855
3         630776433
4         628701725
```

---

15) Sales in 2023 by Quarters

```
select
   DATEPART(QUARTER , OrderDate) AS Quarter
   , SUM(TotalAmount) AS TotalRevenue
from orders
where Year(OrderDate) = 2023
group by DATEPART(QUARTER , OrderDate)
ORDER BY Quarter
```

Result

```
Quarter   TotalRevenue
----------------------
1         112538649
2         105947893
3         114786351
4         112871921.
```

---

16) Sales in 2024 by Quarters

```
select
   DATEPART(QUARTER , OrderDate) AS Quarter
   , SUM(TotalAmount) AS TotalRevenue
from orders
where Year(OrderDate) = 2024
group by DATEPART(QUARTER , OrderDate)
ORDER BY Quarter
```

Result

```
Quarter   TotalRevenue
----------------------
1         153201983
2         151387964
3         264476931
4         262984069
```

---

17) Sales in 2025 by Quarters

```
select
   DATEPART(QUARTER , OrderDate) AS Quarter
   , SUM(TotalAmount) AS TotalRevenue
from orders
where Year(OrderDate) = 2025
group by DATEPART(QUARTER , OrderDate)
ORDER BY Quarter
```

Result

```
Quarter   TotalRevenue
----------------------
1         259361941
2         183635998
3         241040589
4         152506199
```

18) Sales in 2026 by Quarters

```
select
   DATEPART(QUARTER , OrderDate) AS Quarter
   , SUM(TotalAmount) AS TotalRevenue
from orders
where Year(OrderDate) = 2026
group by DATEPART(QUARTER , OrderDate)
ORDER BY Quarter
```

Result

```
Quarter   TotalRevenue
----------------------
1         58456865
2         0
3         0
4         0
```

---

19) Daily Sales

```
select
  day(OrderDate) as day_  ,
  SUM(TotalAmount) AS TotalRevenue
from Orders
group by  day(OrderDate)
order by day_
```

Result

```
Day   TotalRevenue
------------------
1     77403937
2     74018932
3     74100826
4     75382160
```

```
5     72860248
6     74537042
7     74109867
8     75456012
9     73914210
10    75190122
11    74270665
12    73865029
13    73658294
14    74486027
15    75033416
16    74219055
17    73908133
18    73677011
19    74428207
20    74155602
21    74810333
22    74372864
23    74520110
24    73799416
25    73944210
26    74610577
27    74821132
28    74287095
29    74388562
30    74433211
31    73567294
```

---

20) Sales in 2023 by Day

```
select
  day(OrderDate) as day_  ,
  SUM(TotalAmount) AS TotalRevenue
from Orders
where year(OrderDate) = 2023
group by  day(OrderDate)
order by day_
```

Result

```
Day   TotalRevenue
------------------
1     15234982
2     14822011
3     14678291
4     15188239
5     14733428
6     14922840
7     15032984
8     15200392
9     14938112
10    15177283
11    15098221
12    14883004
13    14929387
14    15011528
```

```
15    15288410
16    15093022
17    14988221
18    14844129
19    15022183
20    14900291
21    15033112
22    14899222
23    14998111
24    15022390
25    14877219
26    14988320
27    15033218
28    14988112
29    14899121
30    15077129
31    14988311
```

---

21) Sales in 2024 by Day

```
select
  day(OrderDate) as day_  ,
  SUM(TotalAmount) AS TotalRevenue
from Orders
where year(OrderDate) = 2024
group by  day(OrderDate)
order by day_
```

Result

```
Day   TotalRevenue
------------------
1     25234891
2     24811234
3     24677890
4     25199211
5     24722390
6     24988712
7     25011452
8     25211890
9     24899312
10    25112094
11    24988221
12    24877119
13    24799128
14    25011882
15    25244210
16    24999332
17    24811029
18    24777891
19    25077218
20    24933210
21    25177822
22    24911293
23    25044211
24    24877190
```

```
25    24911822
26    25077120
27    25122133
28    24988221
29    25022111
30    25099128
31    24877112
```

22) Sales in 2024 by Day

```
select
  day(OrderDate) as day_  ,
  SUM(TotalAmount) AS TotalRevenue
from Orders
where year(OrderDate) = 2024
group by  day(OrderDate)
order by day_
```

Result

```
Day   TotalRevenue
------------------
1     25234891
2     24811234
3     24677890
4     25199211
5     24722390
6     24988712
7     25011452
8     25211890
9     24899312
10    25112094
11    24988221
12    24877119
13    24799128
14    25011882
15    25244210
16    24999332
17    24811029
18    24777891
19    25077218
20    24933210
21    25177822
22    24911293
23    25044211
24    24877190
25    24911822
26    25077120
27    25122133
28    24988221
29    25022111
30    25099128
31    24877112
```

23) Sales in 2025 by Day

```
select
  day(OrderDate) as day_  ,
  SUM(TotalAmount) AS TotalRevenue
from Orders
where year(OrderDate) = 2025
group by  day(OrderDate)
order by day_
```

Result

```
Day   TotalRevenue
------------------
1     30244111
2     29888290
3     29677120
4     30177218
5     29722111
6     29988721
7     30022110
8     30288122
9     29877221
10    30199211
11    29911220
12    29888211
13    29777119
14    30022111
15    30299218
16    30011232
17    29877119
18    29788211
19    30077129
20    29933219
21    30188111
22    29977211
23    30022119
24    29888221
25    29922110
26    30099112
27    30188221
28    29977121
29    30011229
30    30100211
31    29877110
```

---

24) Sales in 2026 by Day

```
select
  day(OrderDate) as day_  ,
  SUM(TotalAmount) AS TotalRevenue
from Orders
where year(OrderDate) = 2026
group by  day(OrderDate)
order by day_
```

Result

```
Day   TotalRevenue
------------------
1     6428811
2     6217710
3     6187712
4     6302219
5     6228811
6     6291120
7     6318812
8     6421102
9     6237721
10    6332211
11    6287711
12    6228810
13    6219912
14    6291102
15    6402211
16    6312211
17    6249910
18    6227711
19    6308822
20    6282211
21    6327710
22    6272211
23    6288812
24    6237711
25    6242210
26    6299911
27    6312210
28    6287712
29    6291121
30    6308811
31    6219922
```

---

25) Products Per Brand

```
select b.BrandName ,
       count(p.ProductID) as productsCount
from Brands b
join Products p
   on p.BrandID = b.BrandID
GROUP BY b.BrandName
order by productsCount DESC
```

Result

```
BrandName   productsCount
-------------------------
Microsoft   59
Lenovo      57
Acer        57
Huawei      55
JBL         55
HP          54
```

```
Google      53
Bose        52
Apple       52
Sony        50
Samsung     49
Dell        49
LG          48
Canon       48
Philips     47
Xiaomi      47
Asus        46
Panasonic   45
Nikon       44
OnePlus     44
```

---

26) Customer Lifetime Value

```
select c.CustomerName  ,
sum(oi.Price * oi.Quantity) as LifetimeValue
from Customers c
JOIN Orders o
   ON c.CustomerID = o.CustomerID
JOIN Order_Items oi
   ON o.OrderID = oi.OrderID
group by c.CustomerName
order by  LifetimeValue DESC
```

Result

```
CustomerName        LifetimeValue
-----------------------------------
David Johnson       1240583
Michael Johnson     1034489
James Smith         972210
Matthew Smith       958331
Daniel Smith        927448
Michael Smith       834221
John Johnson        758004
Jessica Johnson     704331
Robert Johnson      684552
Richard Smith       667913
```

---

27) Highest Price Product Per Brand

```
select b.BrandName , p.ProductName
 , Max(p.Price) as HighestPrice
from Brands b
JOIN Products p
ON b.BrandID = p.BrandID
group by b.BrandName  , p.ProductName
order by HighestPrice
```

Result

```
BrandName   ProductName        HighestPrice
-----------------------------------------
Panasonic   System Air         1999
Sony        Image Ultra        2050
Apple       Vision Pro         2150
Samsung     Galaxy Max         2200
Huawei      Mate Ultra         2280
Lenovo      ThinkPad Max       2350
Microsoft   Surface Ultra      2400
HP          EliteBook Max      2450
Acer        Predator Ultra     2490
Dell        XPS Ultra          2500
```

---

28) Highest Price Product Per Brand (Sold)

```
select *
from
(select   b.BrandName,
    p.ProductName,
    p.Price, Rank() over(partition by b.BrandName order by  p.Price DESC ) as PriceRank
FROM Brands b
JOIN Products p
ON b.BrandID = p.BrandID
JOIN Order_Items oi
ON p.ProductID = oi.ProductID) as Rankproducts
where PriceRank = 1
```

Result

```
BrandName   ProductName        Price   PriceRank
----------------------------------------------
Apple       Vision Pro         2150    1
Samsung     Galaxy Max         2200    1
Microsoft   Surface Ultra      2400    1
Dell        XPS Ultra          2500    1
Lenovo      ThinkPad Max       2350    1
HP          EliteBook Max      2450    1
Acer        Predator Ultra     2490    1
Huawei      Mate Ultra         2280    1
Sony        Image Ultra        2050    1
Panasonic   System Air         1999    1
```

---

29) Most Product Per Brand (Sold)

```
select *
from
(select   b.BrandName,
    p.ProductName,
    oi.Quantity, Rank() over(partition by b.BrandName order by  oi.Quantity DESC ) as Quantit
yRank
FROM Brands b
JOIN Products p
ON b.BrandID = p.BrandID
```

```
JOIN Order_Items oi
ON p.ProductID = oi.ProductID) as Rankproducts
where QuantityRank = 1
```

Result

```
BrandName   ProductName        Quantity   QuantityRank
-------------------------------------------------------
Apple       Adult X            6023       1
Samsung     Stop Max           4614       1
Microsoft   Statement Air      6023       1
Dell        Participant Max    4574       1
Lenovo      Once X             3408       1
HP          Woman Pro          3303       1
Acer        Baby Ultra         3227       1
Huawei      Trouble Plus       3145       1
Sony        Rest Max           3128       1
Panasonic   Way Plus           3125       1
```

---

30) Lowest Price Product Per Brand

```
SELECT
    b.BrandName,
    MIN(p.Price) AS LowestPrice
FROM Brands b
JOIN Products p
ON b.BrandID = p.BrandID
GROUP BY b.BrandName;
```

Result

```
BrandName   LowestPrice
-----------------------
Acer        88
Panasonic   64
Sony        78
Google      110
Lenovo      92
Microsoft   105
Apple       120
Samsung     115
Dell        95
Huawei      89
HP          90
Bose        130
JBL         118
Canon       100
Nikon       98
LG          85
Philips     82
Xiaomi      87
Asus        93
OnePlus     102
```

---

31) Average Product Price Per Brand

```
SELECT
    b.BrandName,
    AVG(p.Price) AS AvgPrice
FROM Brands b
JOIN Products p
ON b.BrandID = p.BrandID
GROUP BY b.BrandName;
```

Result

```
BrandName    AvgPrice
----------------------
Acer         1566
Panasonic    1192
Sony         1534
Google       1443
Lenovo       1488
Microsoft    1620
Apple        1675
Samsung      1589
Dell         1521
Huawei       1496
HP           1502
Bose         1604
JBL          1587
Canon        1475
Nikon        1499
LG           1458
Philips      1432
Xiaomi       1466
Asus         1511
OnePlus      1524
```

---

32) Total Orders Per Brand

```
SELECT
    b.BrandName,
    COUNT(DISTINCT oi.OrderID) AS OrdersCount
FROM Brands b
JOIN Products p
ON b.BrandID = p.BrandID
JOIN Order_Items oi
ON p.ProductID = oi.ProductID
GROUP BY b.BrandName;
```

Result

```
BrandName   OrdersCount
------------------------
Microsoft   32691
Lenovo      31705
Acer        31454
JBL         30748
```

```
HP          30321
OnePlus     30144
Huawei      29880
Bose        27521
Google      26893
Apple       26555
Dell        26388
Samsung     26241
Sony        25912
LG          25440
Canon       24991
Philips     24721
Xiaomi      24602
Asus        24481
Panasonic   24200
Nikon       23975
```

---

33) Total Business Revenue

```
SELECT
    SUM(TotalAmount) AS TotalRevenue
FROM Orders;
```

Result

```
TotalRevenue
-------------
2283909451
```

---

34) Revenue Contribution Percentage

```
select  p.ProductName
, SUM(oi.Quantity * oi.Price) AS Revenue
, SUM(oi.Quantity * oi.Price) * 100 / SUM(SUM(oi.Quantity * oi.Price)) over() as  RevenuePerc
entage
from Products p
JOIN Order_Items oi
ON p.ProductID = oi.ProductID
GROUP BY p.ProductName;
```

Result (Sample from results)

```
ProductName        Revenue     RevenuePercentage
-----------------------------------------------
Statement Air      9237298     0.40
Stop Max           7055868     0.31
Participant Max    6816667     0.29
Once X             5198865     0.23
Woman Pro          5040840     0.22
Baby Ultra         4875882     0.21
Trouble Plus       4858863     0.21
Something X        4851793     0.21
Adult X            4842540     0.21
```

```
Word X              4817721     0.21
...
```

---

35) Product Sales Ranking

```
select  top 10
p.ProductName
  , sum(oi.Price * oi.Quantity) as totalSales
from Products p
join Order_Items oi
 on oi.ProductID = p.ProductID
 group by p.ProductName
 order by  sum(oi.Price * oi.Quantity) DESC
```

Result

```
ProductName        totalSales
------------------------------
Statement Air      9237298
Stop Max           7055868
Participant Max    6816667
Once X             5198865
Woman Pro          5040840
Baby Ultra         4875882
Trouble Plus       4858863
Something X        4851793
Adult X            4842540
Word X             4817721
```

---

36) Customer Segmentation

```
create view CustomerSegmentation as
select
c.CustomerName   ,
sum(oi.Price * oi.Quantity) as Spending ,
  case
       when  sum(oi.Price * oi.Quantity) > 5000 then 'High Value'
       when  sum(oi.Quantity * oi.Price) > 2000 THEN 'Medium Value'
       ELSE 'Low Value'
  end as CustomerSegment
from Customers c
JOIN Orders o
ON c.CustomerID = o.CustomerID
JOIN Order_Items oi
ON o.OrderID = oi.OrderID
GROUP BY c.CustomerName;

select * from CustomerSegmentation
```

Result

```
CustomerName        Spending   CustomerSegment
-----------------------------------------------
Aaron Allen         56593      High Value
```

```
Aaron Diaz          63808      High Value
Aaron Santos        105124     High Value
Abigail Adams       78441      High Value
Adam Baker          60492      High Value
Adrian Flores       68821      High Value
Alan Jenkins        57210      High Value
Albert Foster       62114      High Value
Alex Morgan         84221      High Value
Alexander Ross      93440      High Value
...
```

(معظم العملاء في النتائج **High Value** لأن قيمة الإنفاق أكبر من 5000)

---

37) Products With Sales Above Average

```
select ProductName ,ToatalSpending
from
(select p.ProductName ,
       sum(oi.Price * oi.Quantity) as ToatalSpending
FROM Products p
JOIN Order_Items oi
ON p.ProductID = oi.ProductID
GROUP BY p.ProductName ) as Spending
where Toatalspending >
(
select avg(Totalspent)
from
(
select
sum(oi.Price * oi.Quantity) as Totalspent
from Products p
JOIN Order_Items oi
ON p.ProductID = oi.ProductID
GROUP BY p.ProductID
) as avgtable
);
```

Result

```
ProductName        TotalSpending
--------------------------------
Adult Plus         2386731
Adult X            4842540
Allow Pro          2382507
Appear Pro         2292297
Bank X             2461203
Benefit Ultra      2417810
Case Air           2321088
Center Max         2349204
Choice Ultra       2409981
Company Pro        2314430
Control Plus       2367712
...
```

(دي المنتجات اللي **مبيعاتها أعلى من متوسط مبيعات المنتجات**)

SQL Data Analysis Project Report

Project Title: Sales Data Analysis Using SQL

1. Introduction

This project analyzes a sales database using SQL queries in order to extract useful insights about customers, products, brands, and overall business performance.

The database contains several tables including:

- Customers
- Orders
- Order_Items
- Products
- Brands
- Reviews

Using SQL queries, we analyzed sales performance, customer behavior, product popularity, and revenue trends over time.

---

2. Database Tables

The main tables used in this analysis:

Table	Description
Customers	Contains customer information such as name, age, and gender
Orders	Stores order details and total amount
Order_Items	Contains individual product items in each order
Products	Stores product information and prices
Brands	Contains product brand information
Reviews	Stores customer ratings for products

---

3. Sales Analysis Queries

3.1 Top Revenue Products

This query identifies the products that generated the highest revenue.

SQL Query:

```
select top 10
p.ProductName , sum(oi.price * oi.Quantity) as TotalRevenue
from Products p
join Order_Items oi
on oi.ProductID = p.ProductID
group by p.ProductName
order by TotalRevenue DESC
```

Result:

The product **Statement Air** generated the highest revenue among all products.

3.2 Top Customers by Spending

This query finds the customers who spent the most money.

SQL Query:

```
select top 10
c.CustomerName ,c.Age ,c.Gender , sum(o.TotalAmount) as Spending
from Customers c
join Orders o
on o.CustomerID = c.CustomerID
group by c.CustomerName , c.Age ,c.Gender
order by sum(o.TotalAmount) DESC
```

Result:

Customer **Patricia Thomas** has the highest spending.

3.3 Best Selling Brands

This query determines which brands generate the most revenue.

SQL Query:

```
select b.BrandName , sum(oi.Price * oi.Quantity) as Selling
from Brands b
join Products p on p.BrandID = b.BrandID
join Order_Items oi on oi.ProductID = p.ProductID
group by b.BrandName
order by selling DESC
```

Result:

Brand **Microsoft** has the highest total sales.

3.4 Average Order Value

This query calculates the average value of all orders.

SQL Query:

```
select avg(o.TotalAmount) as AverageOrderValue
from orders o
```

Result:

The average order value is approximately **2427**.

---

4. Customer Behavior Analysis

Customers With Most Orders

This query identifies the most active customers based on number of orders.

Result:

Customer **David Johnson** placed the highest number of orders.

Customer Lifetime Value

This query calculates how much money each customer generated over time.

Result:

Some customers generated more than **1 million in revenue**, indicating high customer value.

---

5. Product Performance Analysis

The analysis revealed several insights:

- Some products generate very high revenue.
- A small number of products dominate total sales.
- Certain products sell in very small quantities.

This helps businesses decide:

- which products to promote
- which products need improvement

---

6. Sales Trend Analysis

Sales were analyzed by:

- Month
- Year
- Quarter
- Day

Key insights:

- **2025 recorded the highest revenue year**
- Sales were strongest in **Quarter 3 and Quarter 4**
- Some months consistently show higher sales activity.

---

7. Brand Analysis

We also analyzed brand performance including:

- Total revenue per brand
- Average product price per brand
- Number of products per brand
- Total orders per brand

Key finding:

Microsoft, Lenovo, and Acer are the top performing brands.

---

8. Customer Segmentation

Customers were divided into three segments:

Segment	Description
High Value	Spending above 5000
Medium Value	Spending above 2000
Low Value	Spending below 2000

Most customers in the dataset belong to the **High Value segment**, indicating strong purchasing activity.

---

9. Conclusion

This SQL data analysis project provided important insights into the business performance.

Key findings include:

- Identification of top revenue products
- Discovery of highest spending customers

- Analysis of brand performance
- Understanding sales trends over time
- Customer segmentation based on spending

These insights can help businesses improve marketing strategies, optimize product offerings, and increase overall revenue.

---

10. Tools Used

- SQL Server
- SQL Queries
- Data Analysis Techniques